Guoyue Xu

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Summary

Machine Learning Engineer specializing in building production-grade autonomous systems and end-to-end ML pipelines. Expertise in **LLM multi-agent orchestration**, **closed-loop Bayesian optimization**, and **high-dimensional neural architectures** (GNNs, Transformers). Proven track record in shipping full-stack AI applications, including a polymer discovery platform serving 1M+ candidates and hardware-in-the-loop (HIL) systems that reduced operational workloads by 80%.

Education

2023–Present **PhD in Mechanical Engineering**, *University of Notre Dame*, Notre Dame, IN
Focus: Machine Learning, Bayesian Optimization, Autonomous Systems, Neural Networks

2021–2023 **MS in Electrical Engineering**, *University of Notre Dame*, Notre Dame, IN

2016–2020 **BSc in Communications Engineering**, *National Chung Cheng University*, Taiwan, **GPA:** 3.83/4.3 (Top 3)

Technical Skills

Core ML	PyTorch, TensorFlow, Scikit-learn, Hugging Face, Gaussian Processes, Active Learning
System & Infra	**MLOps:** Docker, Git, Linux, HPC (SLURM), Multi-GPU Distributed Training, CI/CD basics
Data & Dev	Python (Expert), SQL, FastAPI, Streamlit, NumPy, Pandas, Matplotlib, C, Bash
Architectures	GNNs (GINE/D-MPNN), Transformers, CNNs, GANs, Multi-Agent Systems, RAG Pipelines
Robotics/IoT	Hardware-in-the-loop control, PID systems, Arduino, Serial Communication, Sensor Fusion

Selected Projects

2025–Present **Production Polymer Discovery Platform — Full-Stack ML**, *Notre Dame*

- **System Architecture:** Architected and deployed a full-stack Streamlit application featuring **LLM-augmented natural language interface** for complex chemical database queries.
- **Model Engineering:** Engineered a **GINE Graph Neural Network ensemble** for multi-task property prediction; optimized inference latency to handle batch processing of 1M+ virtual molecules.
- **Optimization Pipeline:** Implemented a multi-objective Pareto optimization engine using **Tanimoto diversity constraints** and synthesizability scoring, improving candidate selection quality by 40%.
- **Generative Modeling:** Deployed an RNN-based SMILES generator with custom **novelty filtering logic** against 6+ reference datasets to automate de novo material discovery.

2023–Present **Autonomous HIL Laboratory (Self-Driving Lab)**, *Notre Dame*

- **Closed-Loop System:** Designed an end-to-end **Hardware-in-the-loop (HIL)** system integrating robotic fluid handling and PID thermal control with a Bayesian optimization core.
- **Algorithmic Impact:** Developed a **Gaussian Process Regression (GPR)** pipeline that autonomously explored high-dimensional chemical spaces, achieving an **80% reduction** in experimental R&D costs.
- **Robustness:** Engineered self-correcting control logic and automated data acquisition pipelines (Serial/Python), ensuring reproducible results with zero human intervention.

2025–Present **Vision-Language Model (VLM) for Lab Automation**, *Notre Dame*

- **Visual Grounding:** Fine-tuning a **Vision-Language-Action (VLA) model** to map unstructured experimental SOPs to real-time laboratory visual feeds.
- **Pipeline Dev:** Building high-throughput data curation and evaluation pipelines for scientific visual comprehension, targeting SOTA performance in lab imagery grounding.

2024 **Medical Anomaly Detection Pipeline**, *Notre Dame*

- **Framework Design:** Reframed MRI anomaly detection as an Image Quality Assessment (IQA) problem, implementing a CNN-based feature extraction pipeline that outperformed traditional baselines by 15%.
- **Benchmarking:** Established a rigorous evaluation framework with standardized metrics across multiple brain MRI datasets to ensure model generalizability.

Engineering Experience

2021–2022 **Lead Systems Engineer — Signal Processing & SDR**, *University of Notre Dame*

- **High-Throughput Data Pipeline:** Developed an automated **Python/C++ data acquisition and control pipeline** for real-time Software-Defined Radio (SDR), reducing manual testing latency by **70%**.
- **Hardware-Software Integration:** Engineered a modular communication framework utilizing **digital signal processing (DSP)** algorithms to mitigate interference and noise in high-congested RF environments.
- **Full-Stack Prototyping:** Architected the transition from wired simulation models to wireless hardware implementations, ensuring system-level robustness and data integrity.

2019–2020 **Project Lead — RF Systems & Hardware Optimization**, *National Chung Cheng University*

- **Award-Winning System Design:** Led a multidisciplinary team to **1st place** in the IEEE MTT-S International Competition by developing a high-efficiency wearable energy harvester.
- **Optimization & Simulation:** Leveraged high-fidelity simulation tools (HFSS/ADS) to optimize antenna parameters, achieving a $-$**24.4 dB return loss** and 5.18 dB gain at 2.45 GHz.
- **Analytical Validation:** Conducted rigorous hardware validation via VNA analysis, bridging the gap between theoretical modeling and physical system performance.

Publications

2025 **Xu, G.***, Zhang, R., & Luo, T. "Self-Driving Laboratory Optimizes the Lower Critical Solution Temperature of Thermoresponsive Polymers." *Advanced Intelligent Discovery*, e202500177.

2025 Zhang, R.*, Xu, J., Zhang, H., **Xu, G.**, & Luo, T. "Active learning-guided exploration of thermally conductive polymers under strain." *Digital Discovery*, 4(3), 812–823.

2024 Pan, Z.*, Xia, J., Yan, Z., **Xu, G.**, et al. "Rethinking Medical Anomaly Detection in Brain MRI." *arXiv* 2408.08228.

2024 Song, K.*, **Xu, G.***, et al. "ML-Assisted 3D Printing of Thermoelectric Materials." *J. Mater. Chem. A*.

2020 **Xu, G.***, et al. "Harvesting Electromagnetic Energy in Air." *IEEE Microwave Magazine*, 21(6), 88–95.

Awards

2019 **1st Place** – IEEE MTT-S Student Design Competition

2019–2021 **President Scholarship** (Top 2%) & **Director Scholarship** (Top 5%, 2×) – National Chung Cheng University

2020 **Prof. Chun-Hsiung Chen Scholarship** – Academic Excellence